

August
2026

PASSWORD CRACKING WITH JTR AND NETWORKWALKS TOOLS

CYBERSECURITY & ETHICAL HACKING PROJECT TASKS

B082-Networkwalks

WEEK 3

PROJECT MODULES: 1 & 2

1. Introduction

This report covers two modules, the first module covers password cracking with John the Ripper (JTR) (W3-PM1), while the second module covers password cracking with Networkwalks tools (W3-PM2).

Many files like PDF, ZIP, and Office documents can be locked with a password. When a file is locked, its password is stored in the form of a hash. A hash is a scrambled value that represents the password. To recover the password, we first take out this hash from the file and then run it through a cracking tool that tries different words until it finds a match.

In this lab tasks (W3-PM1) I used JTR John and JTR Johnny to recover the passwords of the protected PDF files. I also used two free online tools made by Networkwalks, namely, the Hash Calculator and Password Cracker. First used the Hash Calculator to take the hash out of the locked PDF files. Then i used the Password Cracker to find the real password from the hash.

2. Definitions and background

Password cracking is the process of recovering a password from stored data or a protected file. Security professionals use it to test how strong a password is and to show why weak passwords are risky.

John the Ripper (JTR) is a popular password cracking tool used by security professionals to test how strong passwords are. It started as a tool for Unix systems but now works on Windows, Linux, and Mac. It can check many types of password hashes and also unlock password protected files like PDF, ZIP, and Office documents.

Johnny is the graphical version of John the Ripper. It gives a simple point and click screen, so beginners can use JTR without typing long commands. Both tools are widely used in security testing and learning labs to understand password safety.

3. Modules completed & Tools utilized

The modules completed and tools utilized are listed in the table below:

Modules	Tools/resources
W3-PM1	John the Ripper (JTR) (Terminal) & Johnny (GUI) JTR_default_password.txt, Online HashCrack, Notepad, and Windows 11 Laptop)
W3-PM2	Online Networkwalks Hash Calculator & Password Cracker, Notepad, and Windows 11 Laptop

4. Targeted Files

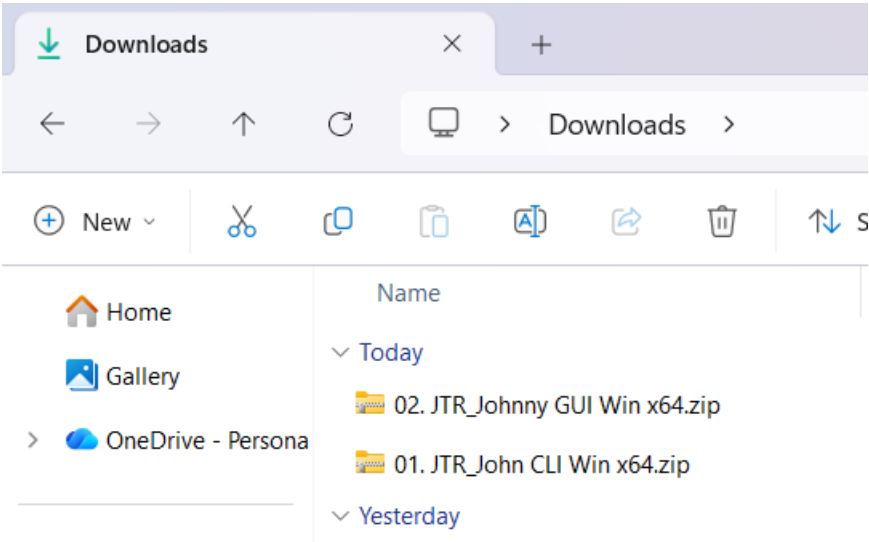
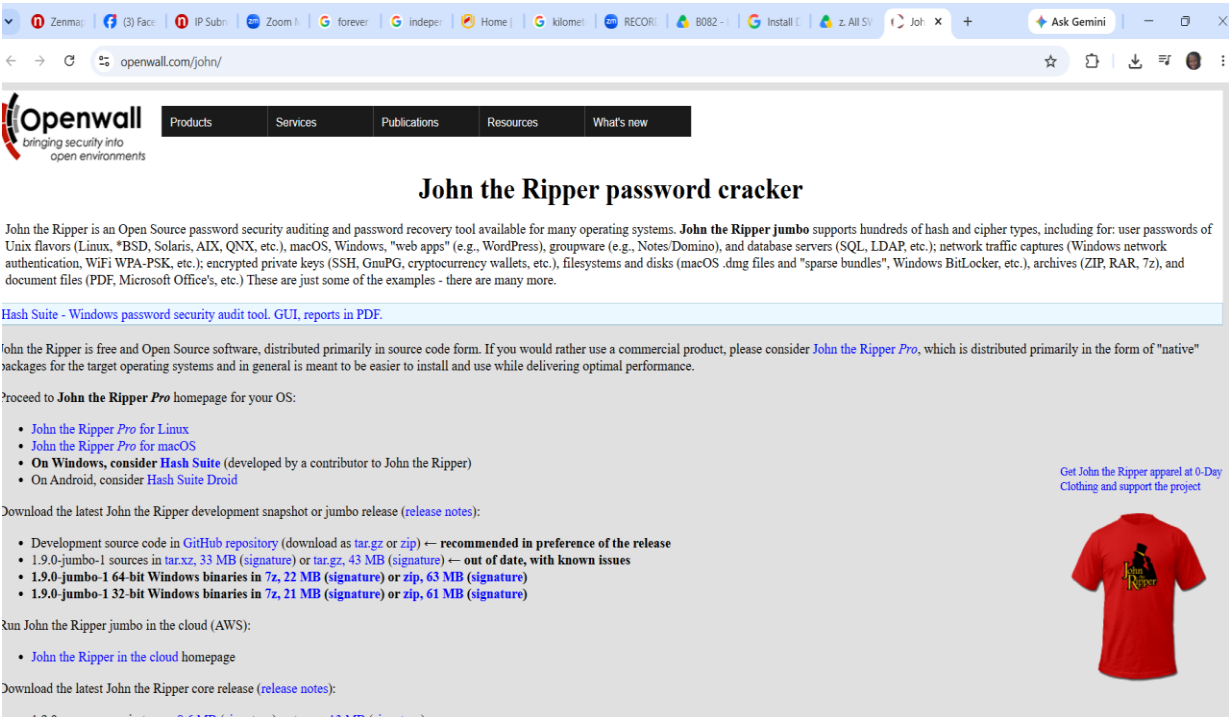
Modules	Password cracked	Status
My Locked PDF1.pdf	good-luck	Password successfully cracked ☒
My Locked PDF2.pdf	password1	Password successfully cracked ☒
My Locked PDF3.pdf	1qaz2wsx	Password successfully cracked ☒

5. Activities Performed

5.1 Module 1: Password cracking with John the Ripper

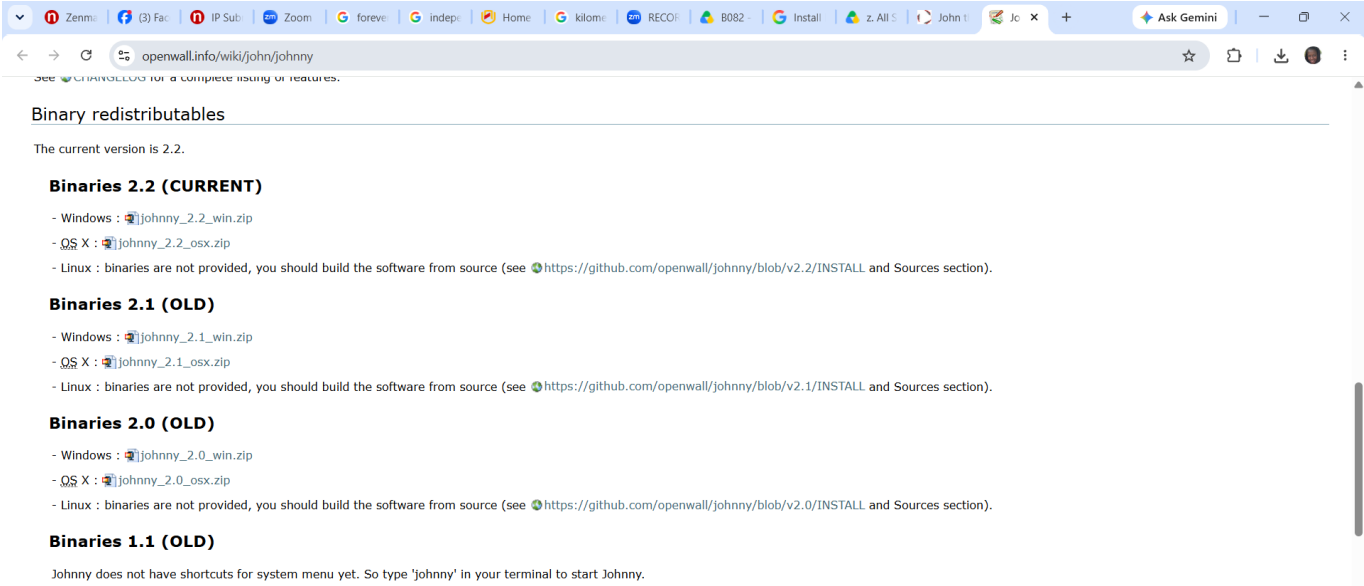
STEP 1

I downloaded John the Ripper from the official website on my windows PC as per the below screenshots.

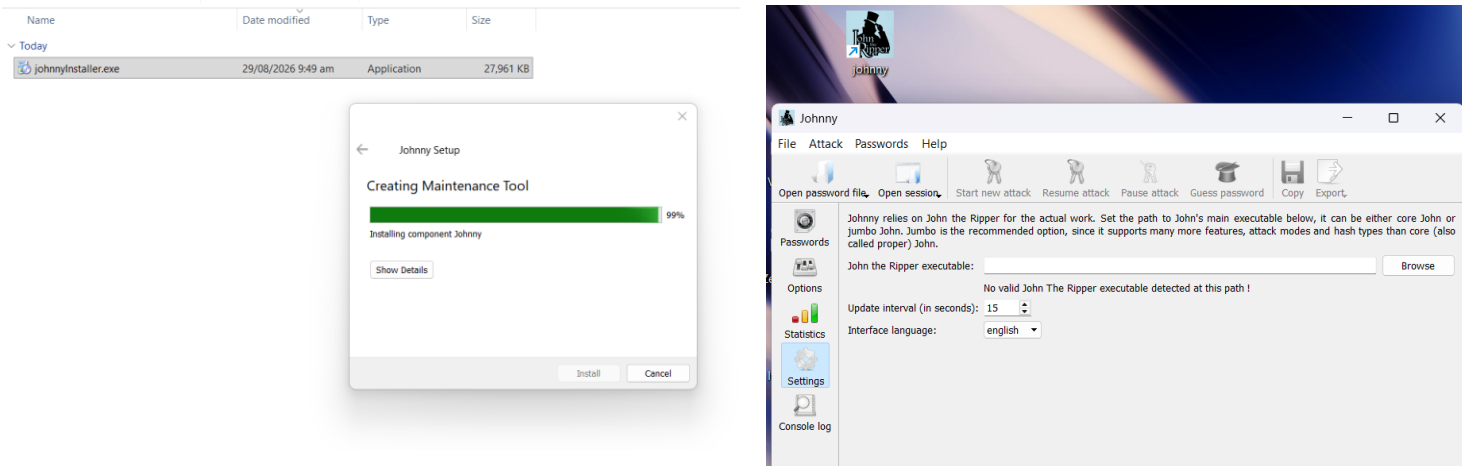


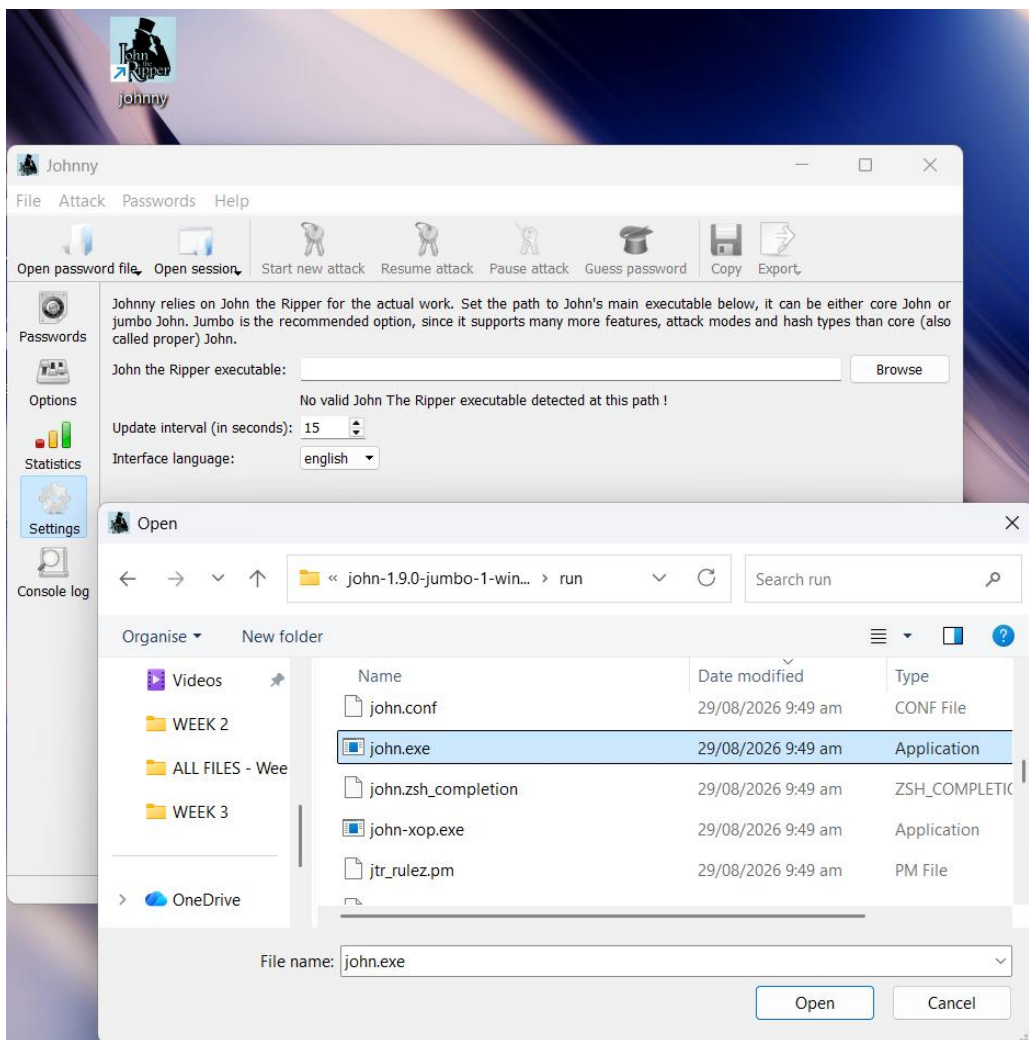
STEP 2

I downloaded and installed Johnny GUI from the official website as shown below:

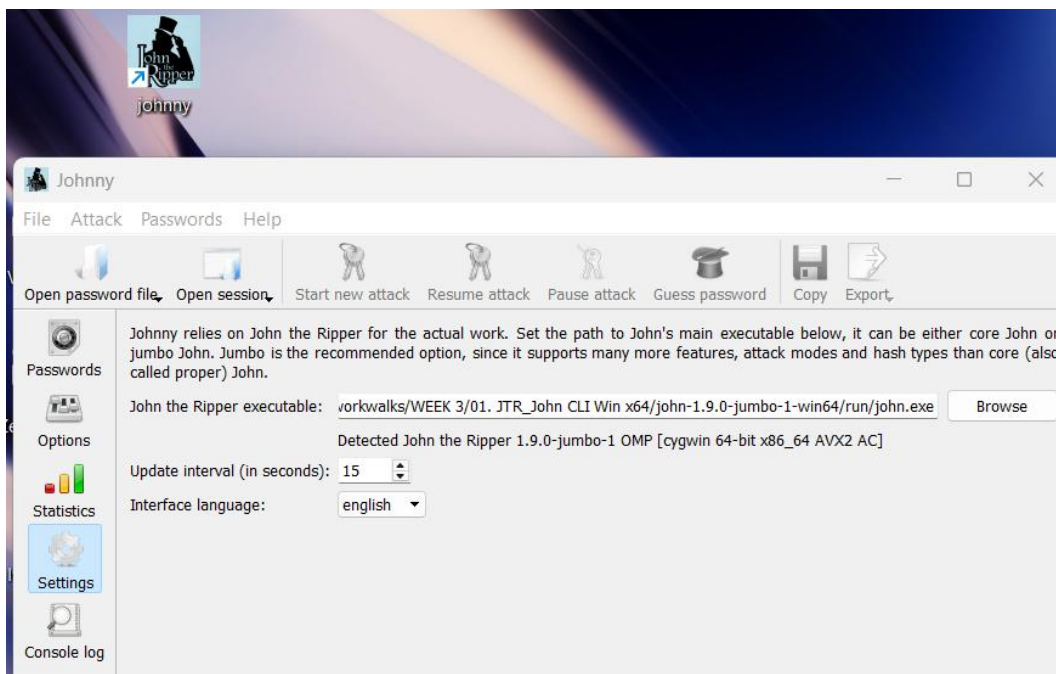


Run the setup file & install Johnny as shown in below:





Select John.exe: *john.exe file is located in the run folder as shown in this screenshot.



STEP 3

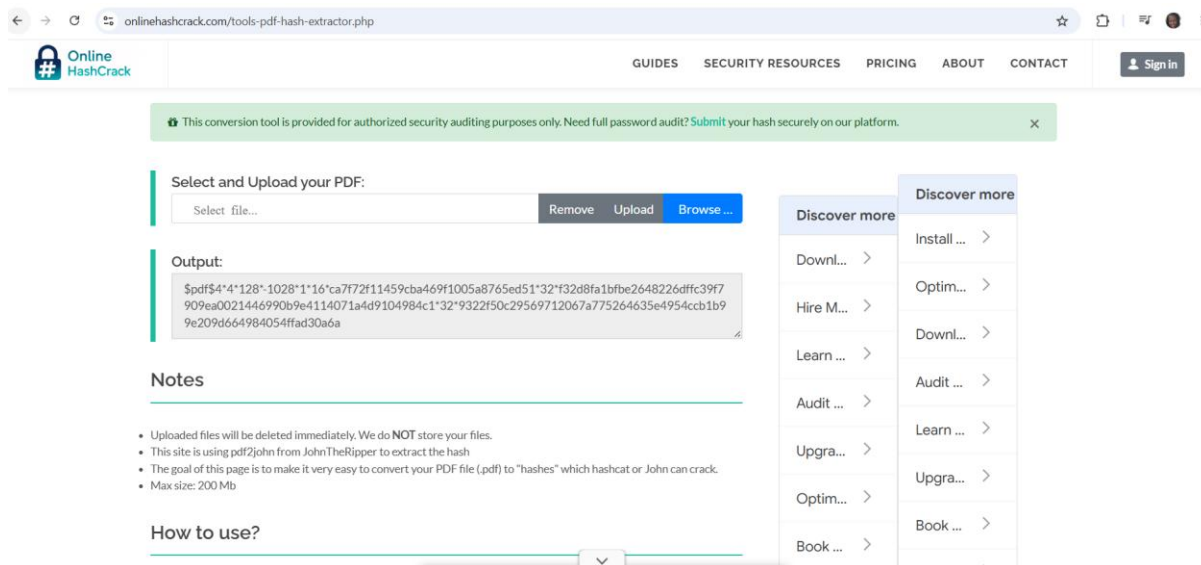
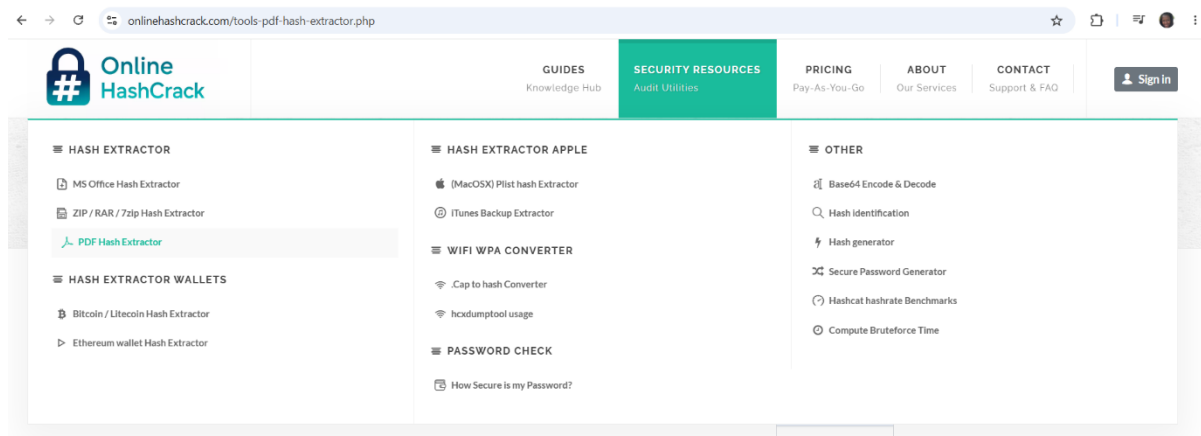
I follow the below steps to crack the password.

I downloaded the encrypted PDF files to my laptop/PC:

Downloads	01. JTR_John CLI Win x64	29/08/2026 9:49 am	File folder	
Pictures	Yesterday			
Documents	My Locked PDF1.pdf	28/08/2026 1:50 pm	Adobe Acrobat D...	267 KB
D:\	My Locked PDF2.pdf	28/08/2026 1:50 pm	Adobe Acrobat D...	205 KB
Music	My Locked PDF3.pdf	28/08/2026 1:50 pm	Adobe Acrobat D...	314 KB
Videos	z. Week3 Projects v1.pdf	28/08/2026 1:49 pm	Adobe Acrobat D...	175 KB

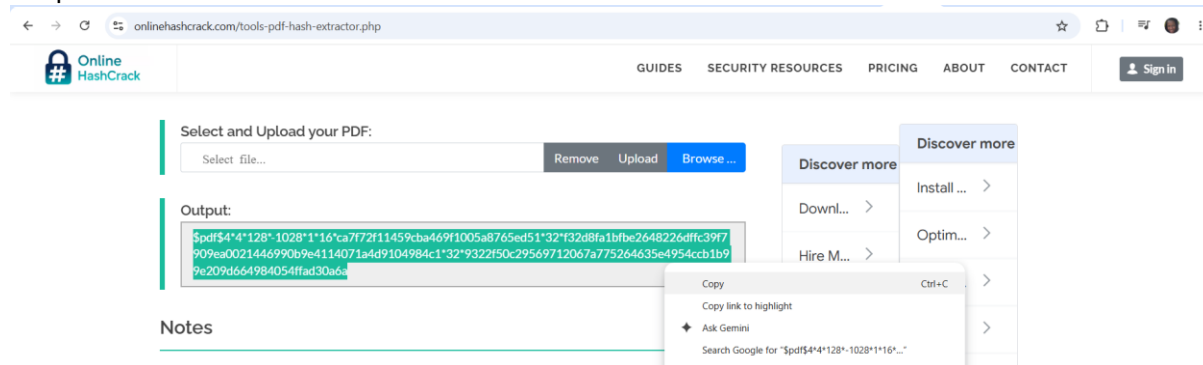
I then opened the hash website and uploaded the pdf files to find the hash:

<https://www.onlinehashcrack.com/tools-pdf-hash-extractor.php>

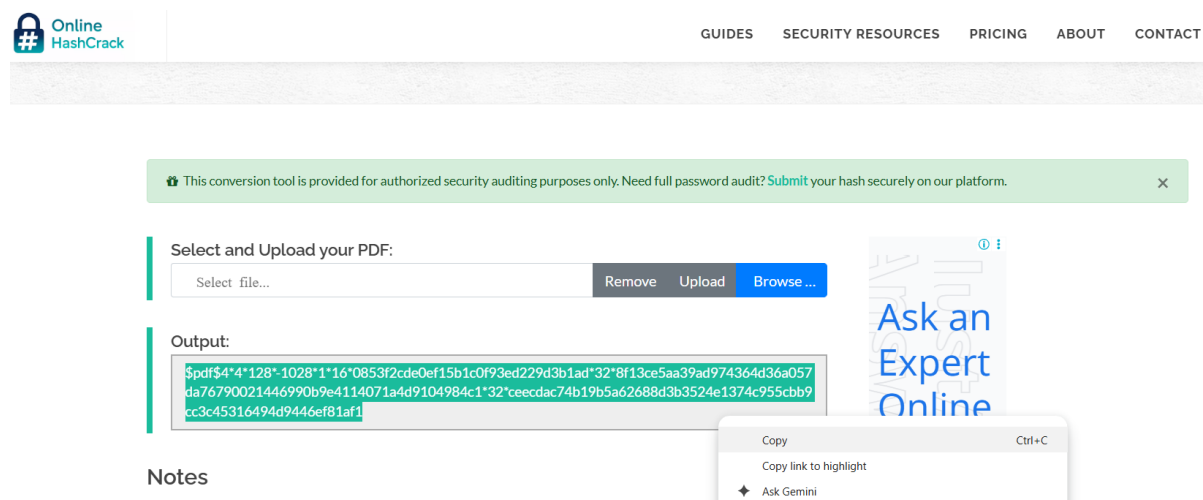


Browse the PDF file and clicked on Upload:

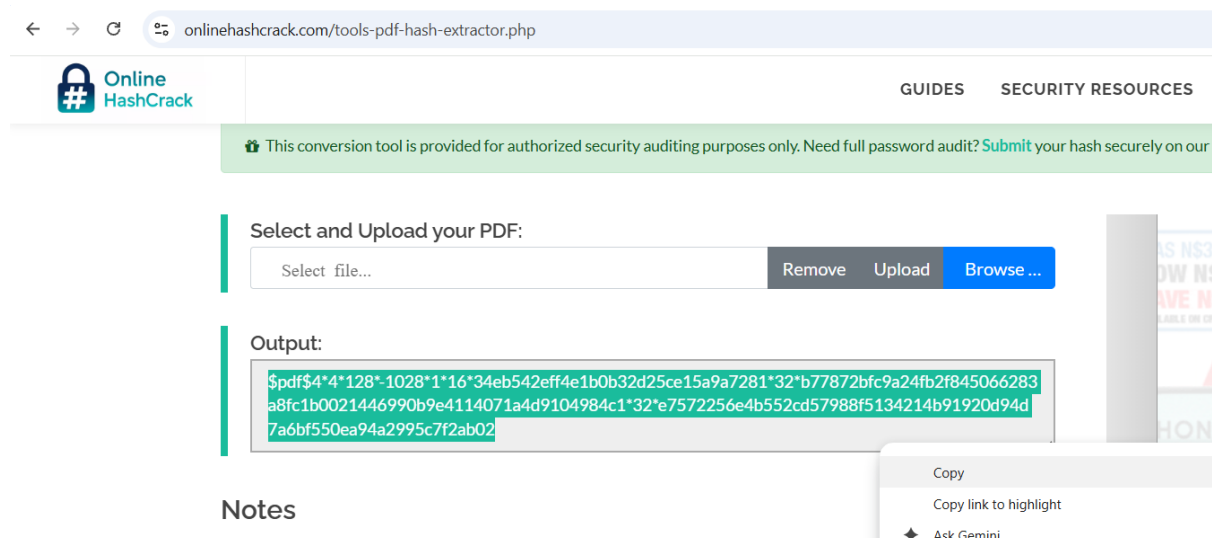
I copied the hash values and thereafter saved them in Note Pad as indicated below:



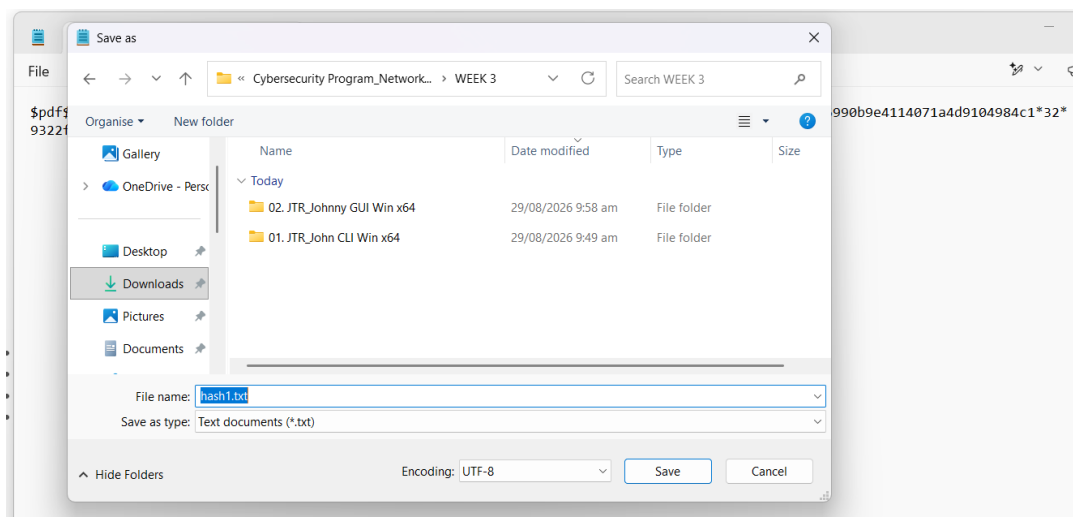
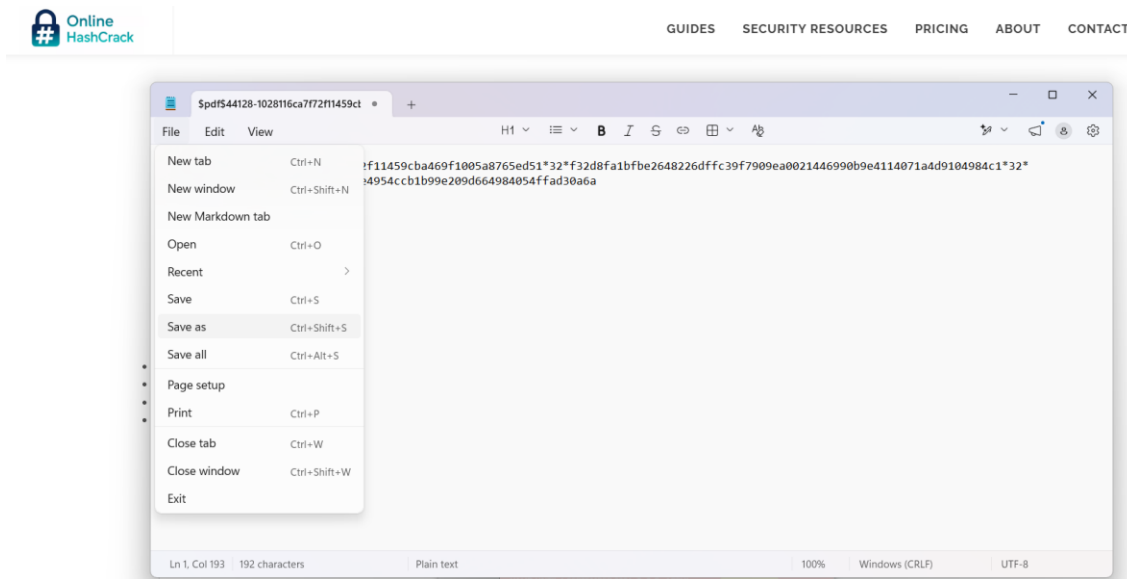
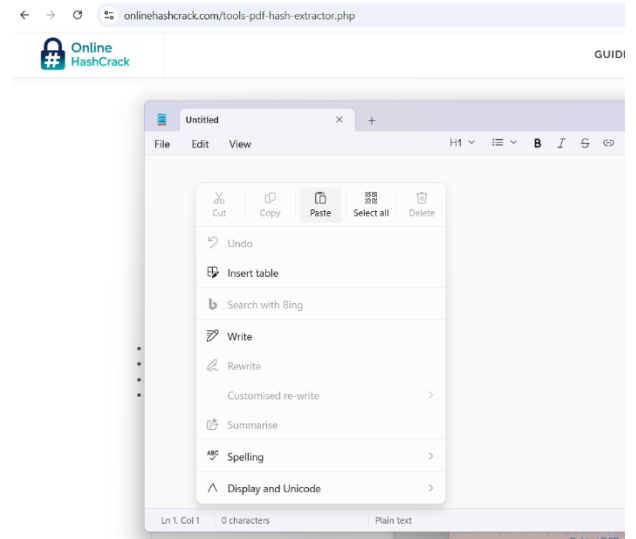
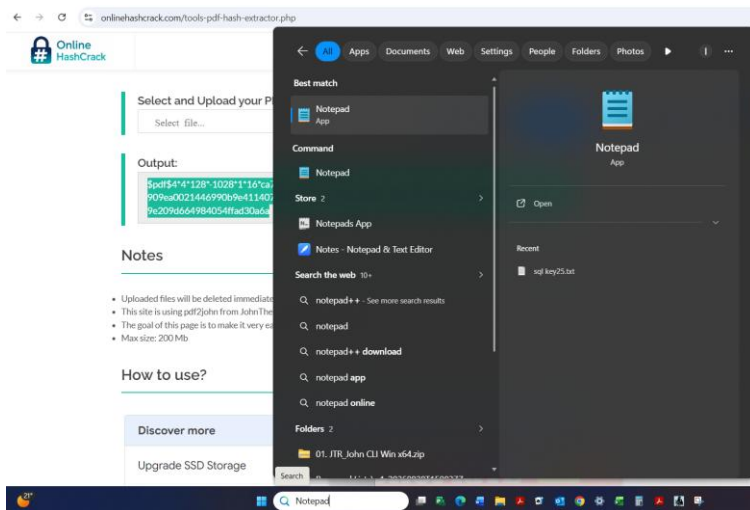
My Locked PDF1 hash value



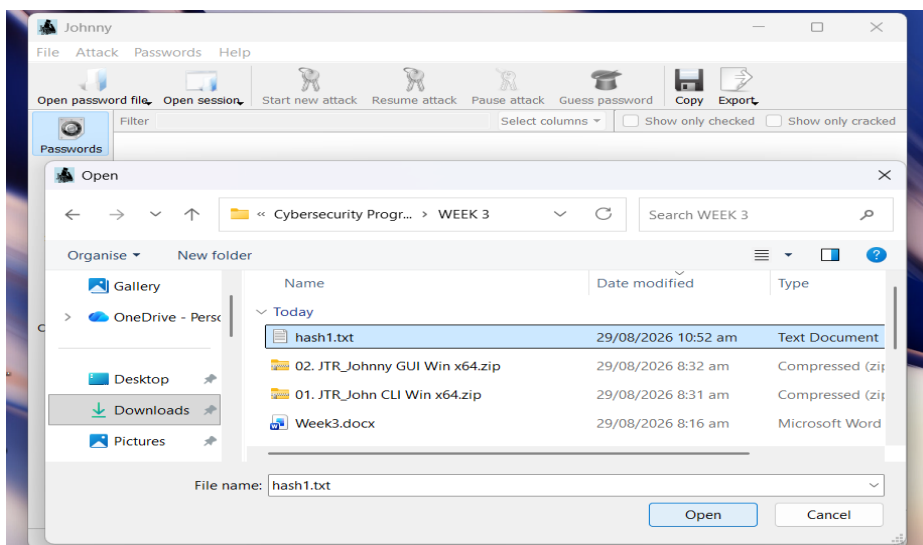
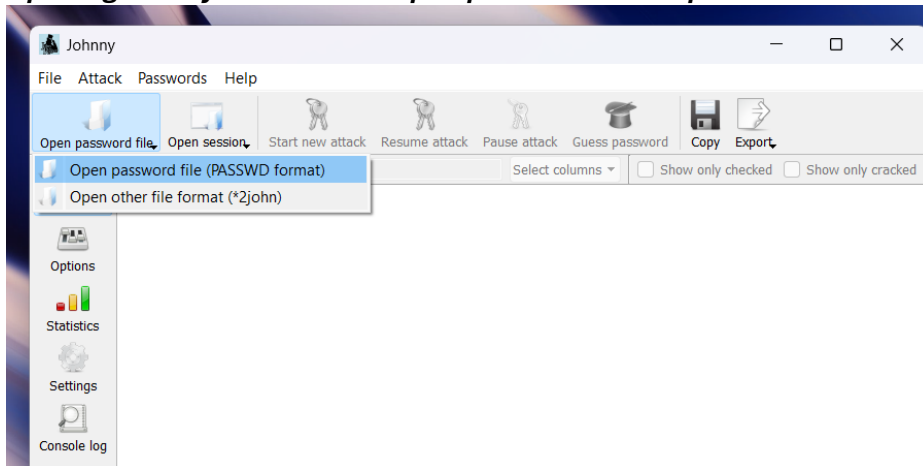
My Locked PDF2 hash value



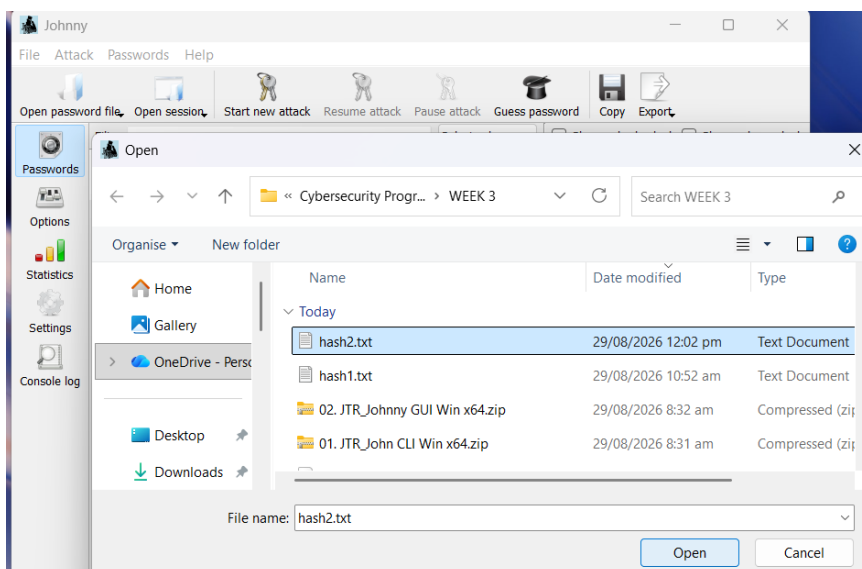
My Locked PDF3 hash value

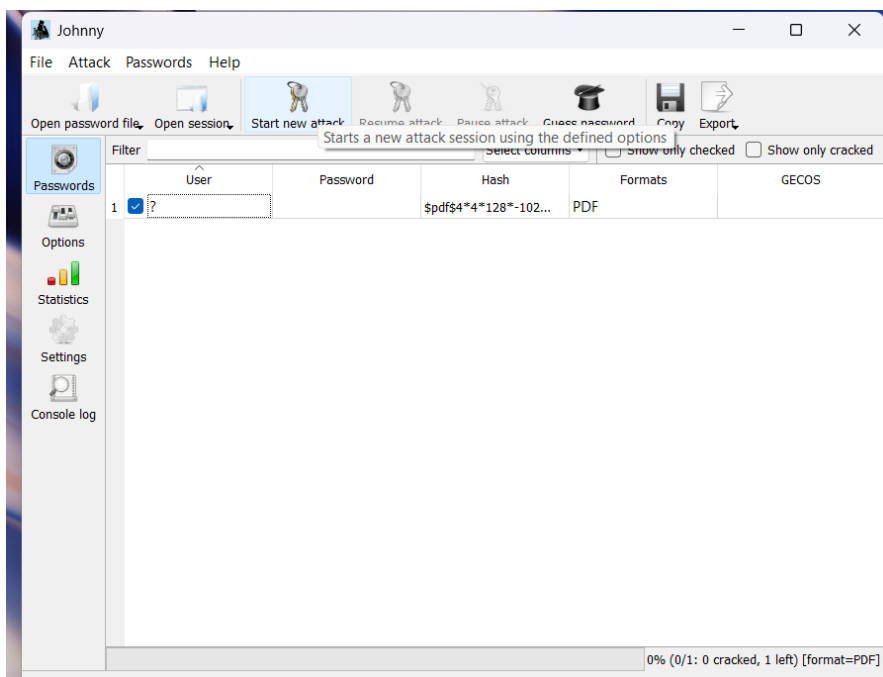
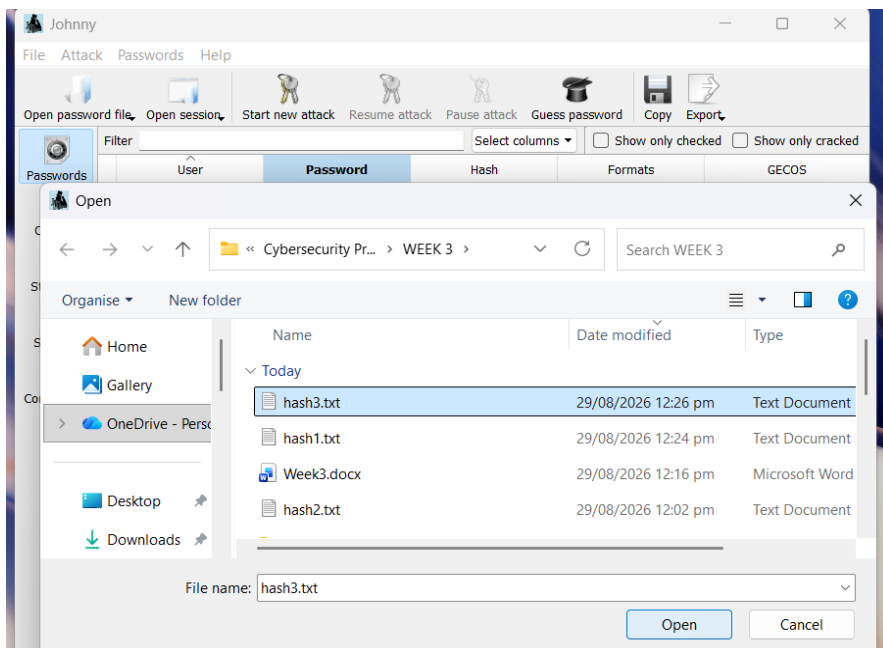


Opening Johnny : i clicked on open password file as per the screenshot below.



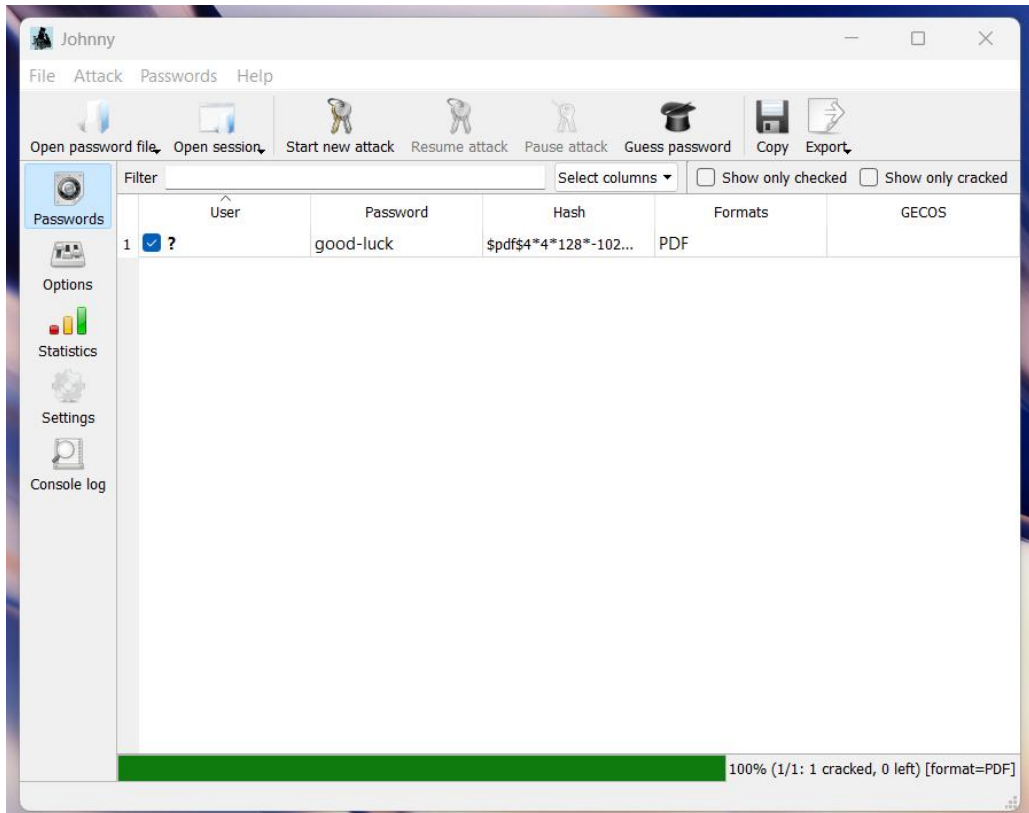
Browsed to the hash .txt file and clicked on open



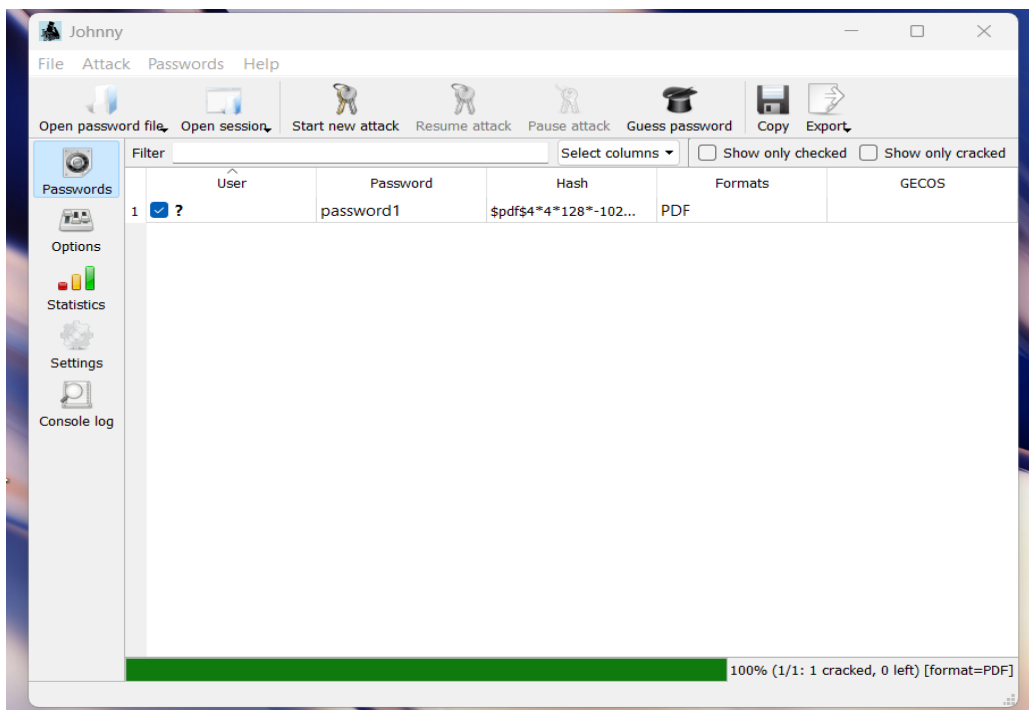


Clicked on 'Start new attack'

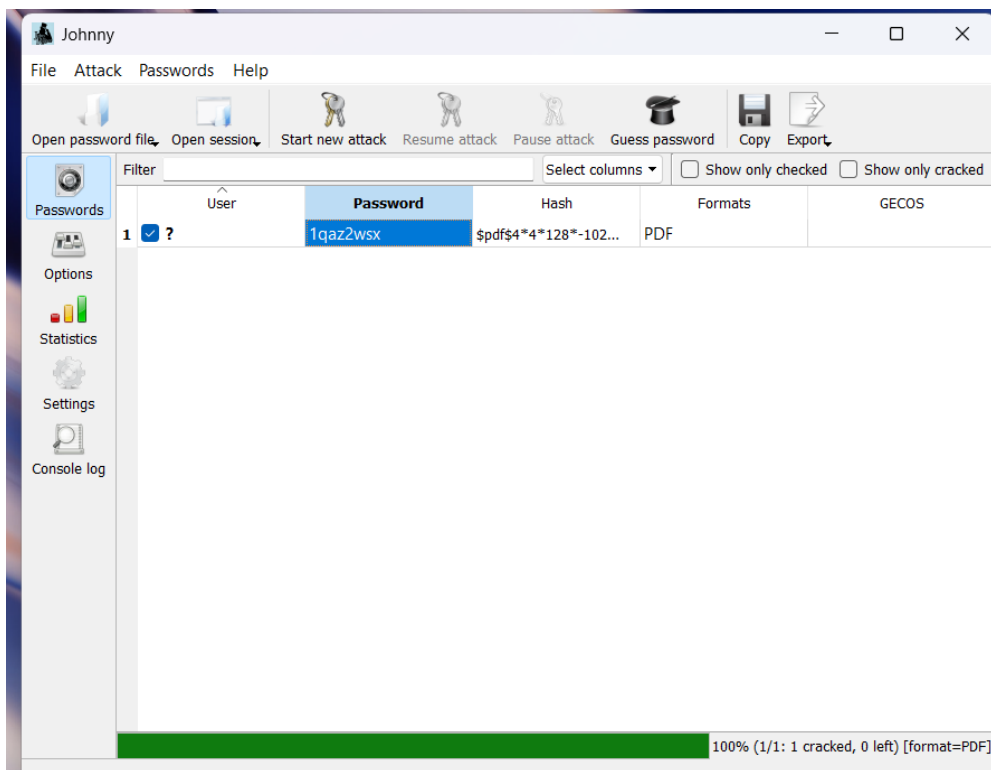
The passwords were cracked and used to open the locked PDF files as per the below screen shots.



My Locked PDF1 password cracked as: *good-luck*

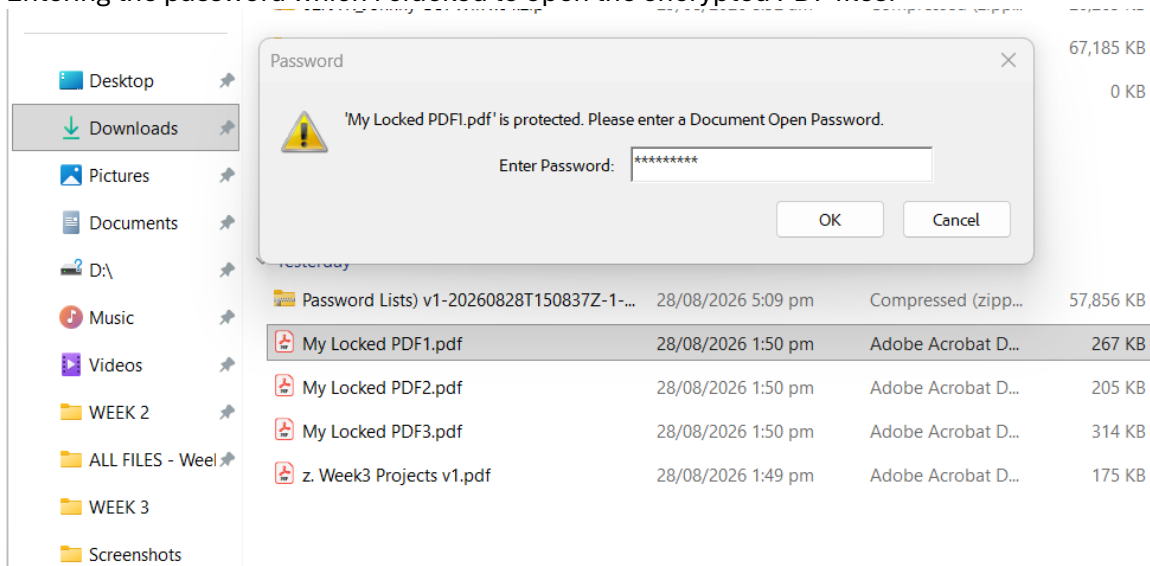


My Locked PDF2 password cracked as: *password1*

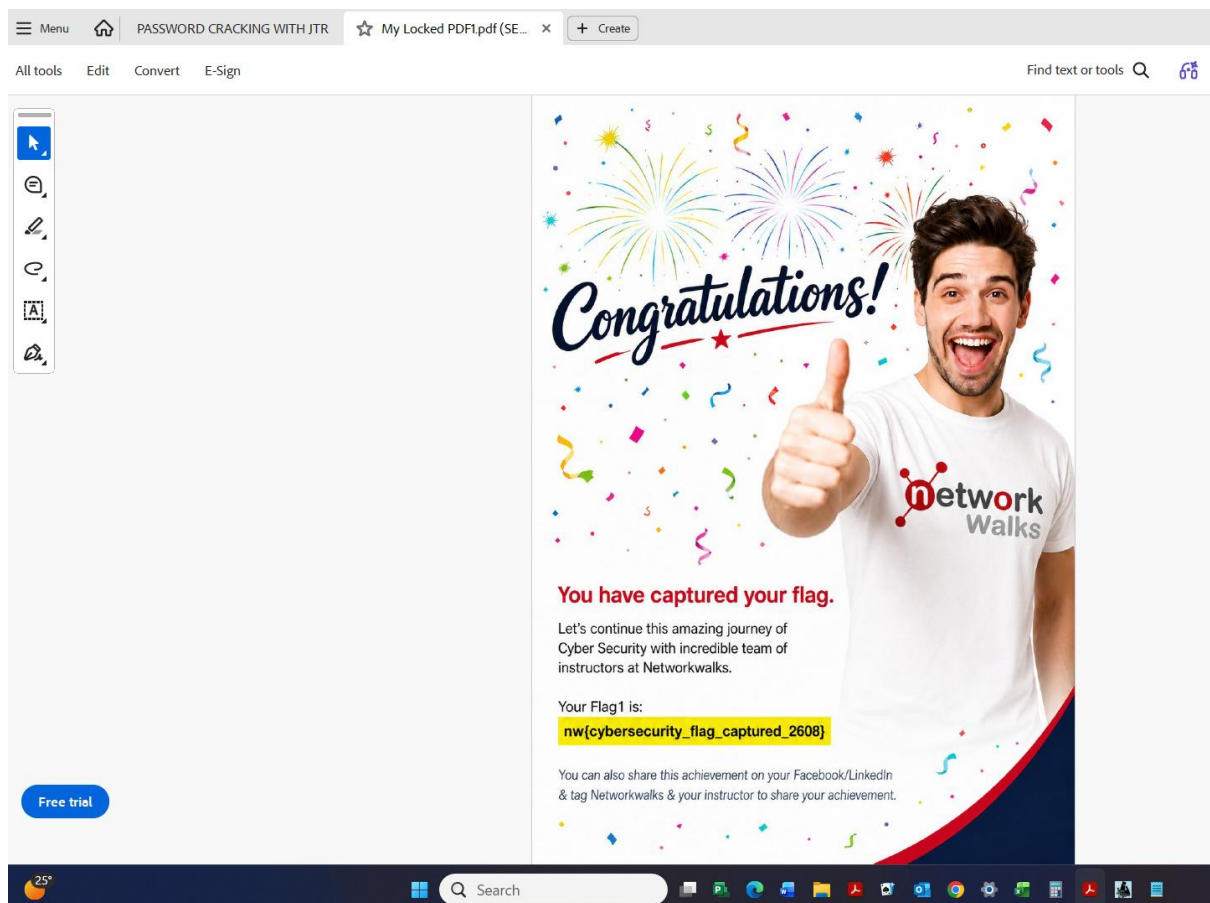


My Locked PDF3 password cracked as: *1qaz2wsx*

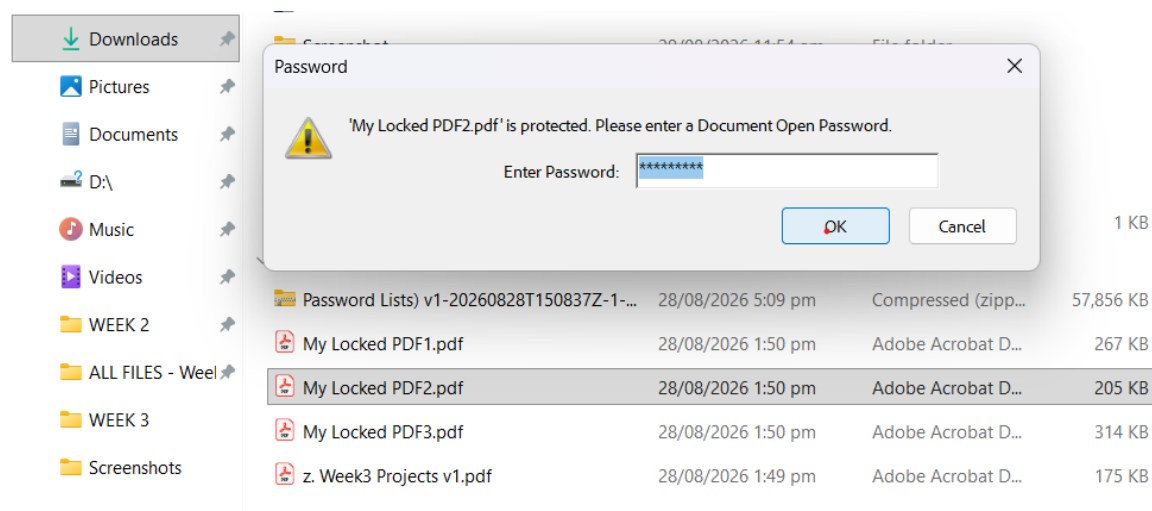
Entering the password which i cracked to open the encrypted PDF files.



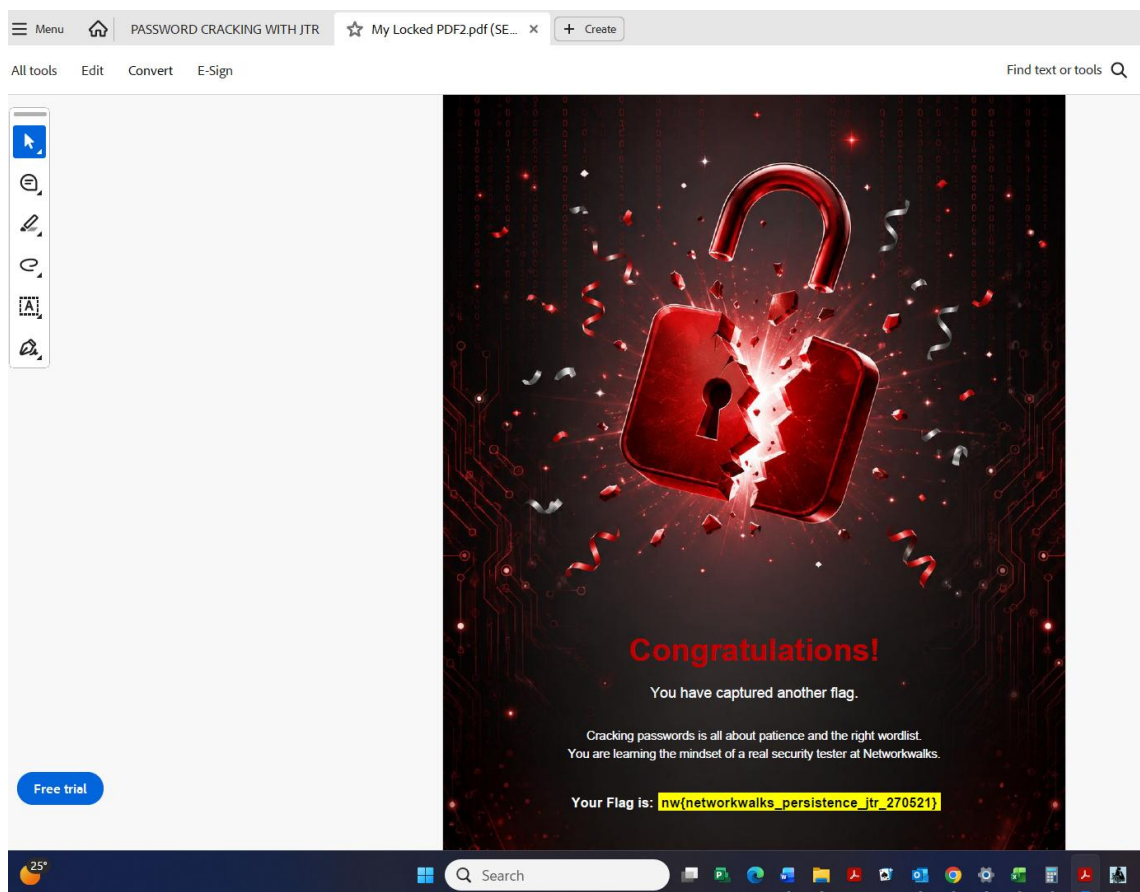
Opening My Locked PDF1



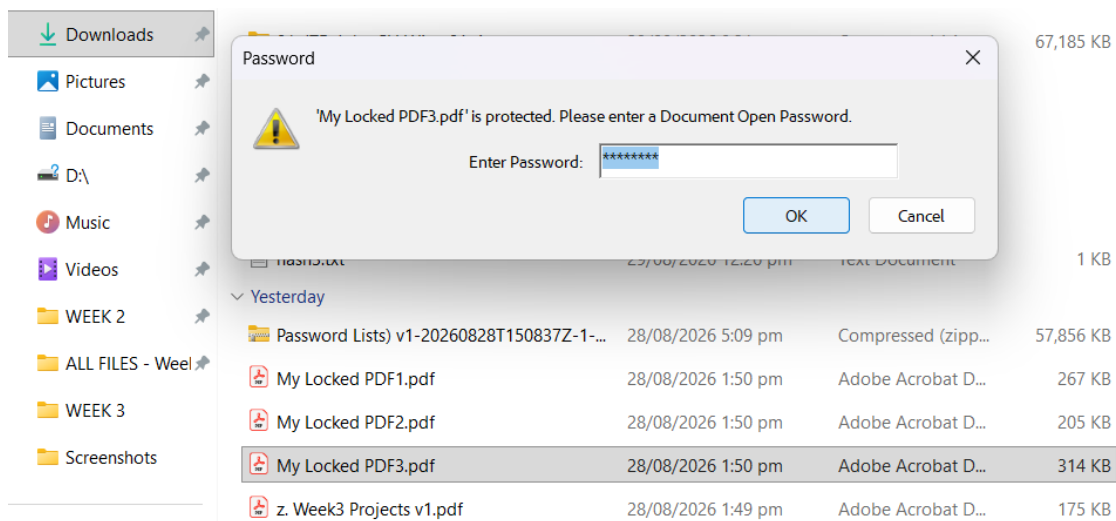
Opened “My Locked PDF1”



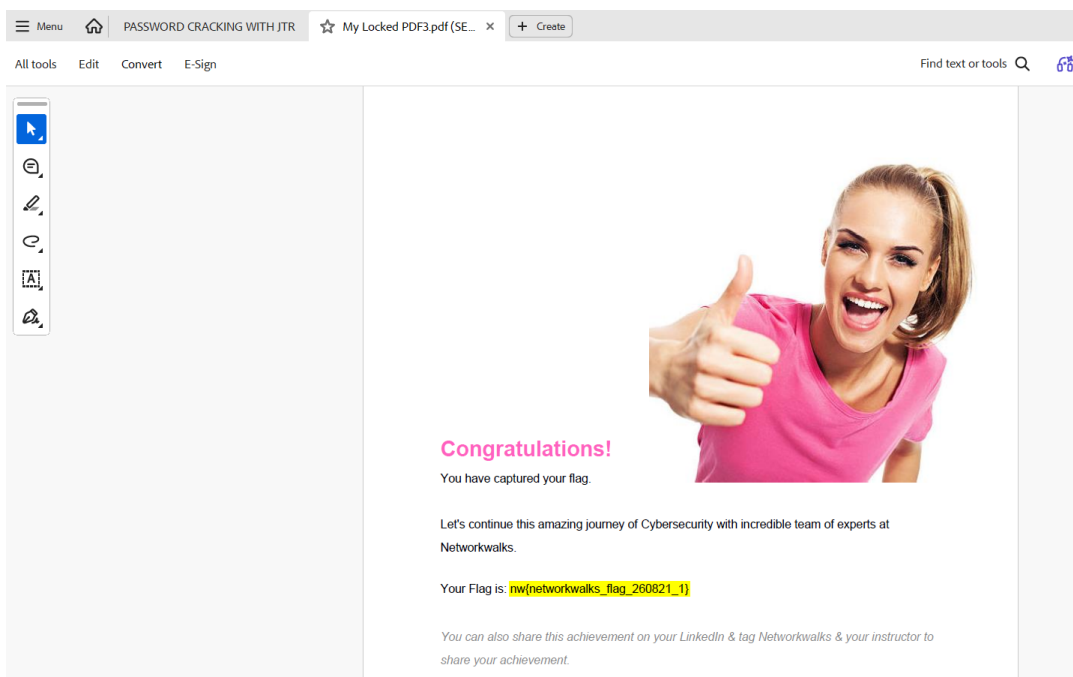
Opening My Locked PDF2



Opened “My Locked PDF2”



Opening My Locked PDF3



Opened “My Locked PDF3”

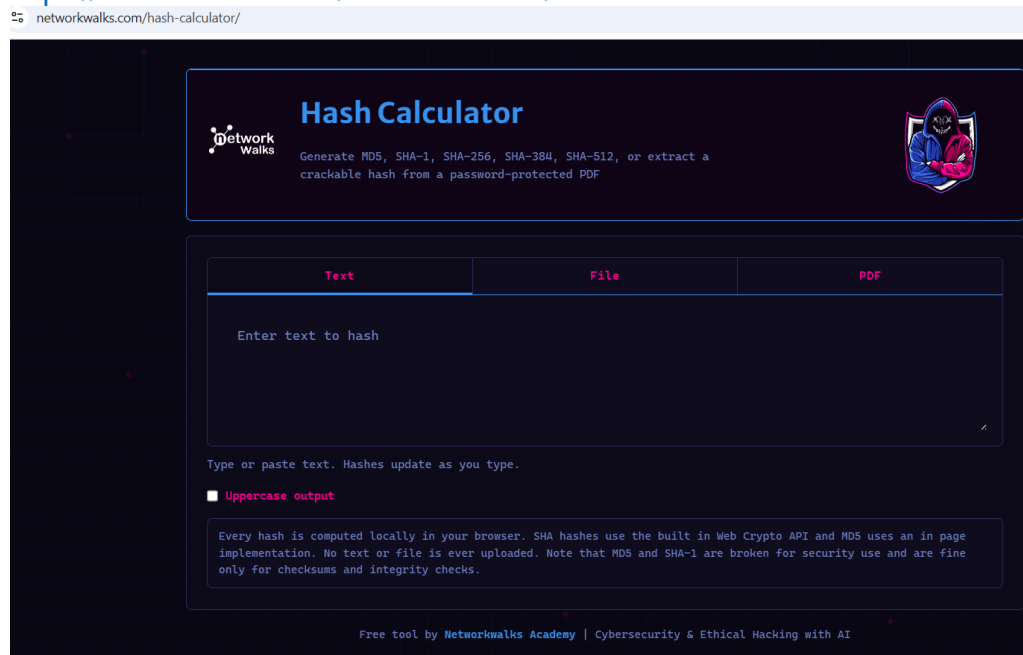
5.2 Module 2: Password cracking with Networkwalks Tools

STEP 1

I Download the encrypted PDF files to my laptop from the lab page using the below link:
<https://networkwalks.com/project-task-lab-password-cracking-with-networkwalks-tools/>

STEP 2

I opened the Networkwalks Hash Calculator in the web browser as per the below:
<https://networkwalks.com/hash-calculator/>



STEP 3

I Upload the locked PDF files to the Hash Calculator. The tool read the files and gave the hash value which started with \$pdf\$...

networkwalks.com/hash-calculator/



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



TextFilePDF



Click to choose a PDF, or drop one here
Nothing is uploaded, the PDF is parsed locally

My Locked PDF1.pdf | 266.9 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

\$pdf\$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe264822
6dfffc39f7909ea0021446990b9e4114071a4d9104984c1*32*9322f50c29569712067a77526463
5e4954ccb1b99e209d664984054ffad30a6a

CopyDownload

Revision: R4 Version: V4 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

STEP 4

I copied the full hash value of all the locked files as indicated below:

My Locked PDF1.pdf | 266.9 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

\$pdf\$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe264822
6dfffc39f7909ea0021446990b9e4114071a4d9104984c1*32*9322f50c29569712067a77526463
5e4954ccb1b99e209d664984054ffad30a6a

CopyDownload

Revision: R4 Version: V4 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

Free tool by Networkwalks Academy | Cybersecurity & Ethical Hacking with AI

My Locked PDF1 hash value;

Text

File

PDF



Click to choose a PDF, or drop one here

Nothing is uploaded, the PDF is parsed locally

My Locked PDF2.pdf | 204.7 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

\$pdf\$4*4*128*-1028*1*16*0853f2cde0ef15b1c0f93ed229d3b1ad*32*8f13ce5aa39ad974364d36a057da76790021446990b9e4114071a4d9104984c1*32*ceecdac74b19b5a62688d3b3524e1374c955cbb9cc3c45316494d9446ef81af1

Copy

Download

Revision: R4 Version: V4 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

My Locked PDF2 hash value;

Text

File

PDF



Click to choose a PDF, or drop one here

Nothing is uploaded, the PDF is parsed locally

My Locked PDF3.pdf | 313.5 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

\$pdf\$4*4*128*-1028*1*16*34eb542eff4e1b0b32d25ce15a9a7281*32*b77872bfc9a24fb2f845066283a8fc1b0021446990b9e4114071a4d9104984c1*32*e7572256e4b552cd57988f5134214b91920d94d7a6bf550ea94a2995c7f2ab02

Copy

Download

Revision: R4 Version: V4 Key length: 128 bit

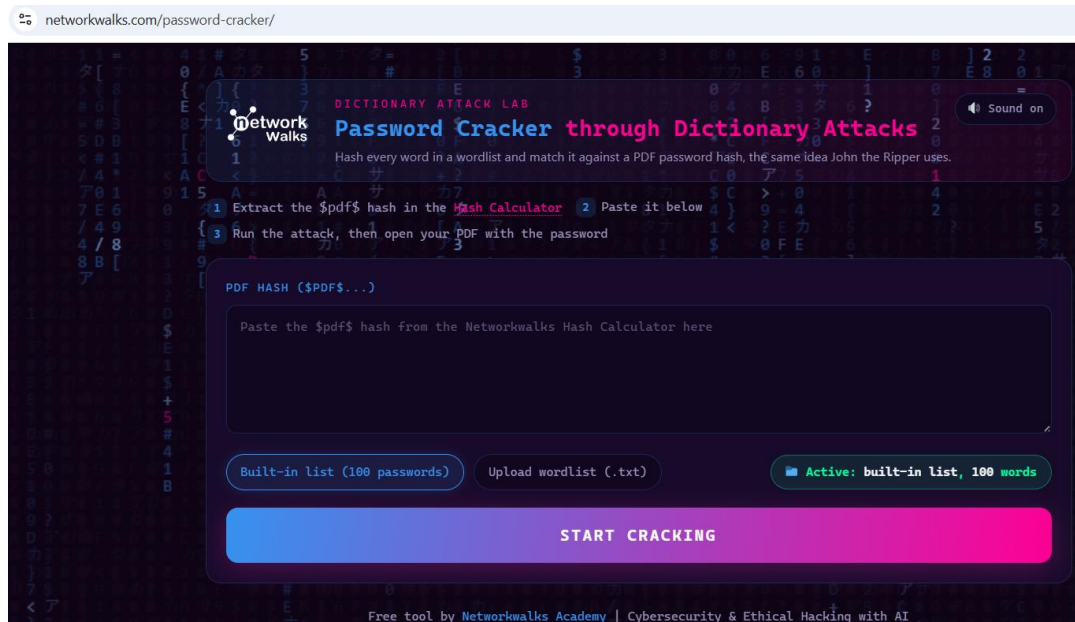
Upload a PDF to check if it is password protected and extract a crackable hash.

My Locked PDF3 hash value;

STEP 5

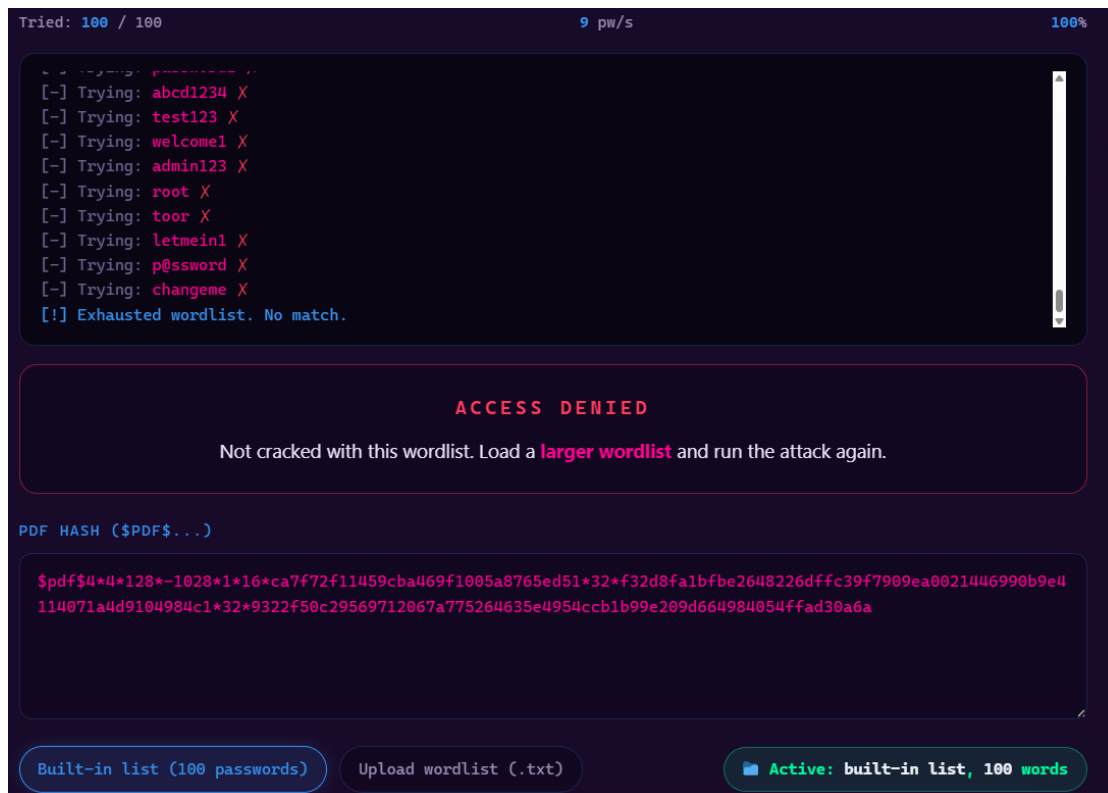
There after i opened the Networkwalks Password Cracker in my web browser using the below link:

<https://networkwalks.com/password-cracker/>

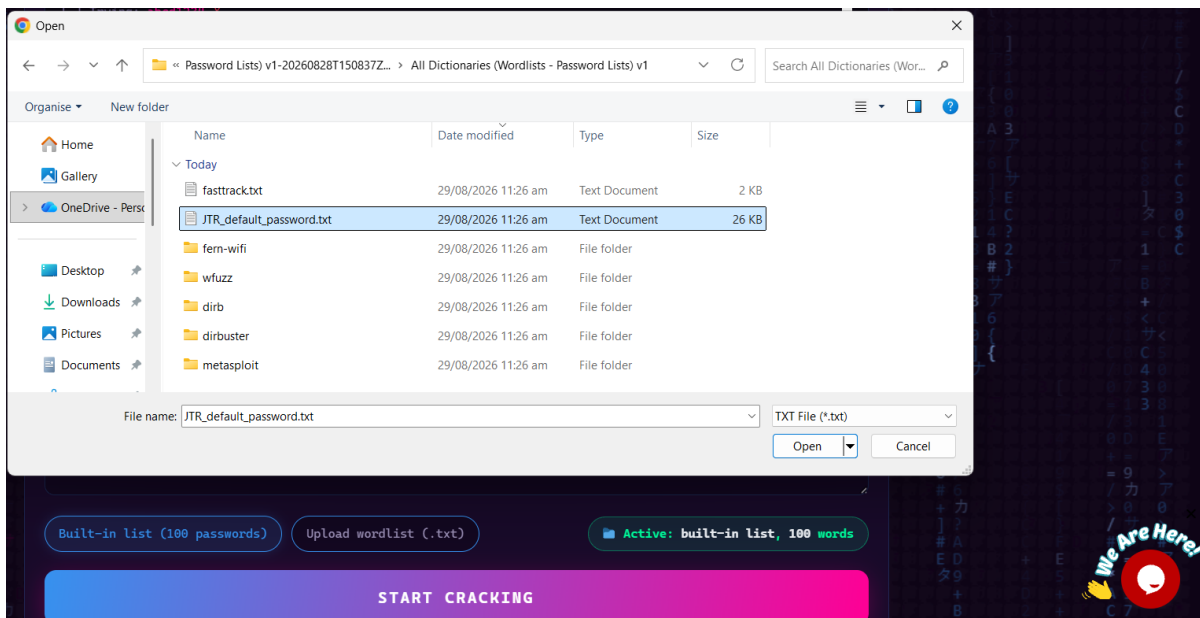


STEP 6

I then pasted the hash value into the Password Cracker and started the attack. The tool tried various passwords until it found the match.



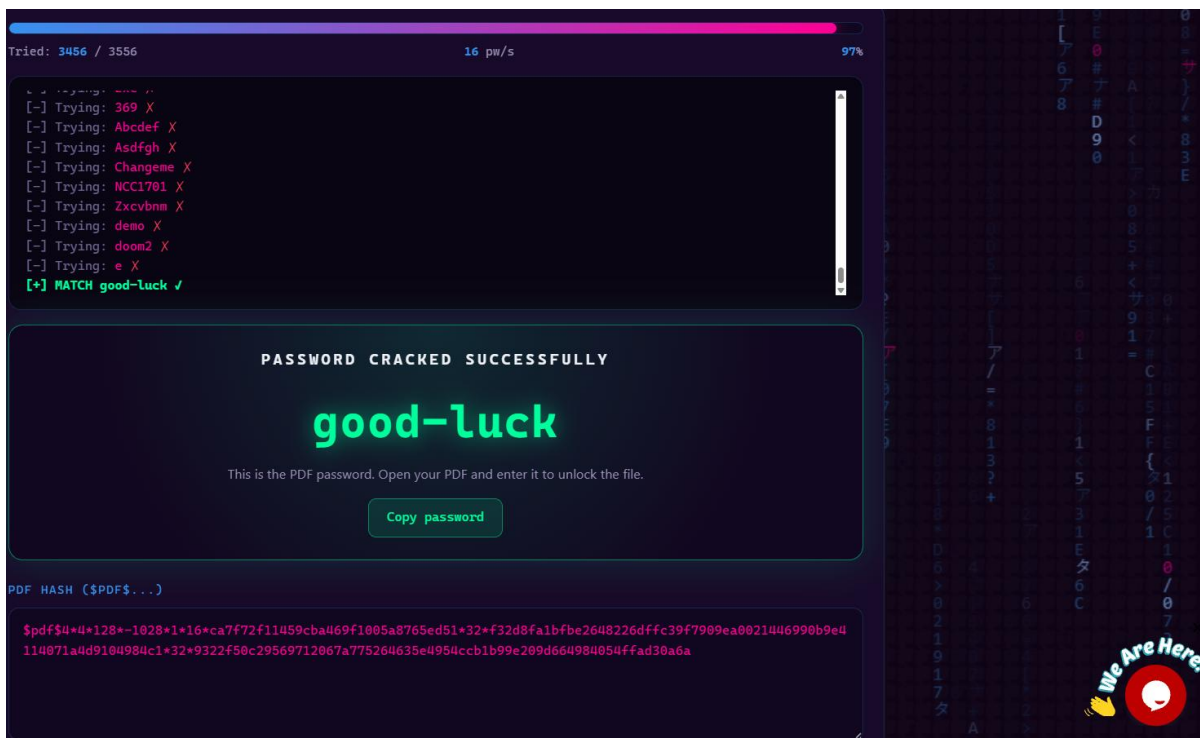
Exhausted the built-in list and no match was found.



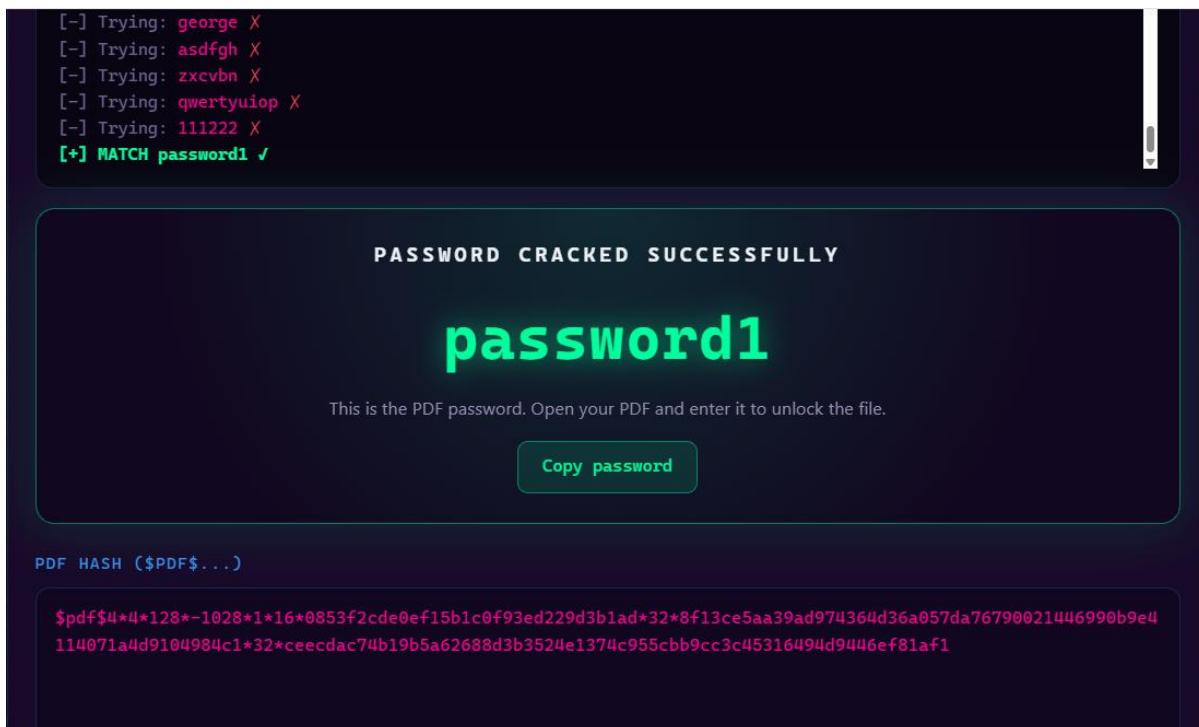
Uploading the JTR default password list.

STEP 7

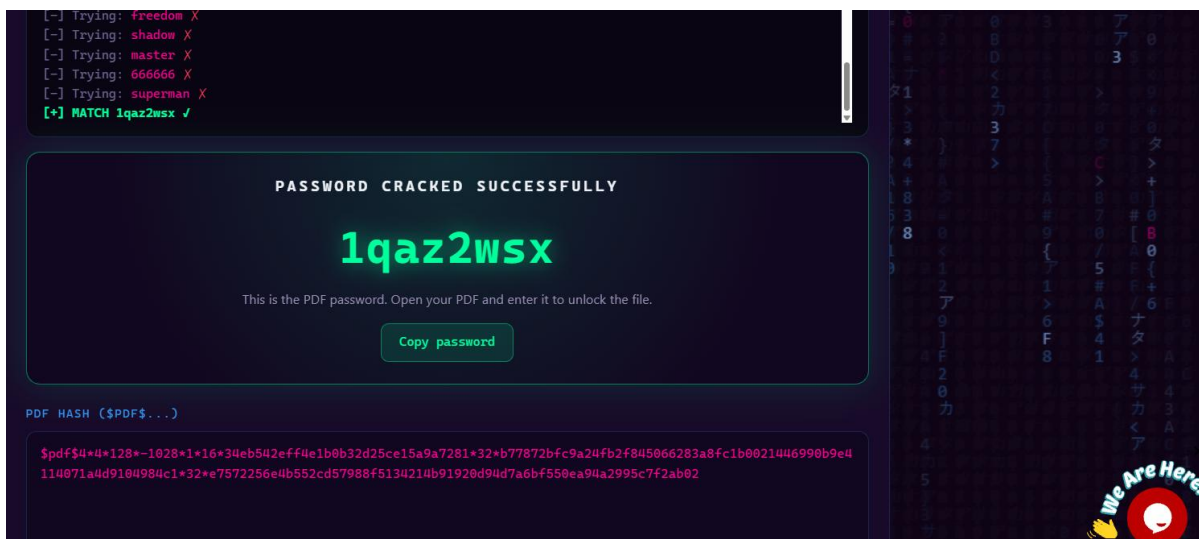
I waited for the tool to finish. The cracked password are indicated in green in the below screenshots.



"My Locked PDF1" password successfully cracked;



“My Locked PDF2” password successfully cracked;



“My Locked PDF3” password successfully cracked;

STEP 8

I open the locked PDF file and enter the cracked password. The results of the opened files were the same as per, 5.1 Module 1: Password cracking with John the Ripper above.

6. Conclusion

This lab helped me understand how password cracking works step by step and why it is important to use strong passwords for protection. I also learned that if a password is short or common, it can be found quickly, which proves the need for strong passwords.

~~~~~END~~~~~



### Author

**Iyaloo Shivute**

Cybersecurity Intern B082

LinkedIn: [www.linkedin.com/in/iyaloo-shivute](https://www.linkedin.com/in/iyaloo-shivute)



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### Project Information

**Program Name:** Cybersecurity program at Networkwalks | **Week:** 03 | **Repository:** GitHub